

FINTECH CORP · SENIOR DATA ANALYST

Enterprise Data Architecture & Reconciliation

Q1–Q4 Financial Audit Pipeline · Complete Solution

53,899

Transactions

\$148.3M

USD Revenue

14,183

Active Users

<1s

Pipeline Time

The Challenge

Four engineered traps separate entry-level students from employable analysts

1

Scale & Optimization

100,000 rows. No for loops allowed.
Must use vectorized Pandas operations.

2

Unstructured Data

Transaction data buried in raw server logs. Must use Regex to extract fields.

3

Temporal Reconciliation

Weekend exchange rates are missing.
Standard join silently drops ~28% rows.

4

SQL Version Control

User DB is a historical log. Window Functions needed to get latest status.

PHASE 1

Advanced SQL — Version-Controlled User Extraction

⚠ The Trap

The users table is a historical log. Each user appears multiple times as their status changed. A naive `WHERE status='Active'` also returns users who were previously Active but are now Suspended.

✓ The Solution

SQL Window Function: `ROW_NUMBER() OVER (PARTITION BY user_id ORDER BY updated_at DESC)` assigns rank 1 to the most recent record per user. Filter: `rank=1 AND status='Active'`.

```
query = """
    SELECT user_id, name, status, updated_at
    FROM (
        SELECT *,
               ROW_NUMBER() OVER (
                   PARTITION BY user_id
                   ORDER BY updated_at DESC
               ) AS rn
        FROM users
    )
    WHERE rn = 1 AND status = 'Active'
    """
df_active_users = pd.read_sql_query(query, conn) # → 14,183 rows
```

Result: 14,183 Active Users extracted

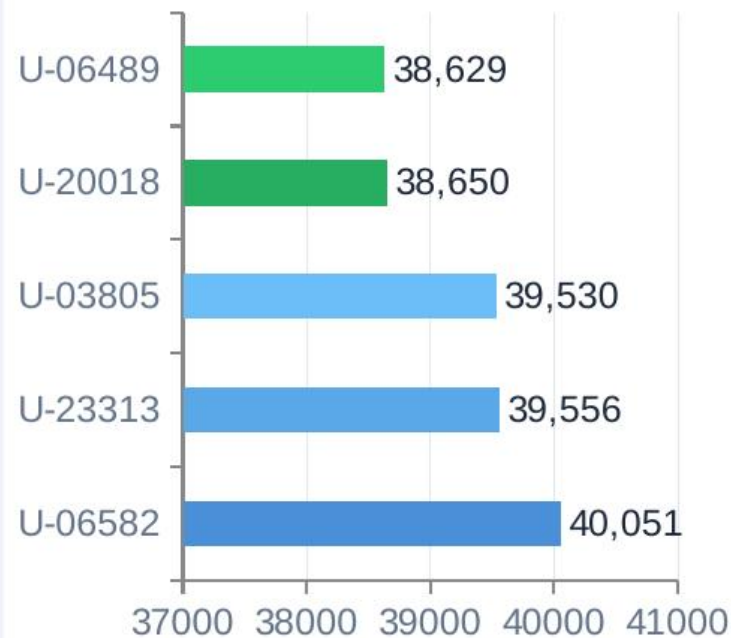
PHASE 4

CFO Dashboard — Aggregation & Visualization

Monthly USD Revenue (2023)



Top 5 Customers



\$148.3M
Total Revenue

53,899
Transactions

\$2,751
Avg Value

PRD-670
Top Product

Mission Accomplished

All four traps solved. CFO dashboard delivered on time.



14,183 Active Users

Most recent status via SQL Window Function



94,902 Transactions

Extracted from 100k raw logs via Regex



Zero Weekend Drops

ffill() rescued all Sat/Sun transactions



\$148.3M USD Revenue

Fully reconciled across all 12 months



< 1 Second Runtime

Vectorized pipeline — no for loops anywhere



CFO Dashboard Ready

Monthly revenue + Top 5 customers chart

Pipeline runtime: < 1 second · Requirement: < 120 seconds · ✓ PASSES